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Scenarios

2026-03-27 04:10:50 PM

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Summary | Tags | Feature: ***../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature*** | **Certification Intelligence Layer — Insights, NL Query, Time-Travel**

insights intelligence ms: 560

Scenario: [1:99] **Identical files produce zero anomalies — insights engine must not hallucinate**

```
>> Background:
   # Most critical quality gate for any intelligence engine.
   # If anomalies are reported when no data changed, the system is fabricating observations.
   # Byte-identical files represent a zero-diff context — any anomaly count above zero fails.
103 Given path '/api/v1/insights'                                                          1
104 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certIdenticalUrl)" }     1
105 When method post                                                                       320
106 Then status 200                                                                        8
107 And match response.anomalies == '#array'                                               30
108 * assert response.anomalies.length == 0                                                16
16:10:49.979 ../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:108
* assert response.anomalies.length == 0
did not evaluate to 'true': response.anomalies.length == 0
../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:108
```

insights intelligence ms: 56

Scenario: [2:111] **Identical files produce zero actionable insights — no fabricated recommendations**

```
>> Background:
   # When there is no diff there can be no actionable recommendations.
   # Fabricating insights from a zero-change dataset is the hallucination equivalent
   # on the insights layer. The array must be present but empty.
115 Given path '/api/v1/insights'                                                          0
116 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certIdenticalUrl)" }     1
117 When method post                                                                       18
118 Then status 200                                                                        0
119 And match response.actionableInsights == '#array'                                      4
120 * assert response.actionableInsights.length == 0                                       1
16:10:50.067 ../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:120
* assert response.actionableInsights.length == 0
did not evaluate to 'true': response.actionableInsights.length == 0
../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:120
```

insights intelligence ms: 73

Scenario: [3:123] **Numeric trend insights carry both absolute and percentage change as real numbers**

```
>> Background:
   # "AWS certifications increased" without a number is noise, not an insight.
   # Every trend entry must carry a calculated absoluteChange AND percentageChange
   # as genuine numbers — null or string values like "N/A" indicate the math was skipped.
127 Given path '/api/v1/insights'                                                          0
128 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)" }      1
129 When method post                                                                       13
130 Then status 200                                                                        0
131 And match response.numericTrends == '#array'                                           3
132 * assert response.numericTrends.length > 0                                             10
133 * def quantified = karate.filter(response.numericTrends, function(t){ return typeof     37
      t.absoluteChange === 'number' && typeof t.percentageChange === 'number' })
134 * assert quantified.length == response.numericTrends.length                            4
```

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2026-03-27 04:10:50 PM

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insights intelligence ms: 35

Scenario: [4:137] **Each actionable insight carries a substantive description grounded in the diff**

```
>> Background:
   # An insight saying "AWS certs grew by 7 (+58%)" can only exist if the engine
   # read the actual row-level diff. Placeholder or single-word descriptions prove
   # the pipeline fired without reading any data.
141 Given path '/api/v1/insights'                                                    0
142 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)" }  0
143 When method post                                                                 14
144 Then status 200                                                                  0
145 And match response.actionableInsights == '#array'                                1
146 * assert response.actionableInsights.length > 0                                  4
147 * def substantive = karate.filter(response.actionableInsights, function(i){ return  3
      i.description && i.description.length > 20 })
148 * assert substantive.length == response.actionableInsights.length               5
```

insights intelligence ms: 54

Scenario: [5:151] **Anomaly count for spike fixture exceeds anomaly count for normal comparison**

```
>> Background:
   # cert_mar2026_spike.xlsx contains a deliberate large spike in Security certifications.
   # An engine reading the actual data must detect this as anomalous.
   # The spike comparison must produce strictly more anomalies than the clean Jan-vs-Mar
   run.
155 Given path '/api/v1/insights'                                                    0
156 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)" }  3
157 When method post                                                                 15
158 Then status 200                                                                  0
159 * def normalAnomalyCount = response.anomalies.length                             3
161 Given path '/api/v1/insights'                                                    0
162 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certSpikeUrl)" }   4
163 When method post                                                                 17
164 Then status 200                                                                  0
165 * def spikeAnomalyCount = response.anomalies.length                               1
166 * assert spikeAnomalyCount > normalAnomalyCount                                   2
```

insights intelligence ms: 35

Scenario: [6:169] **Trend insights reference technology dimension names — not generic column labels**

```
>> Background:
   # Insights must be domain-aware: referencing actual technology names (AWS, Azure, GCP,
   etc.)
   # from the certification data, not opaque labels like "Column B" or "Field 3".
   # Each trend must carry a dimension field that names the domain entity it describes.
173 Given path '/api/v1/insights'                                                    1
174 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)" }  0
175 When method post                                                                 16
176 Then status 200                                                                  0
177 * assert response.numericTrends.length > 0                                       0
178 * def domainAware = karate.filter(response.numericTrends, function(t){ return t.dimension  6
      && t.dimension.length > 1 })
179 * assert domainAware.length > 0                                                  1
```

ask intelligence ms: 38

Scenario: [7:197] **Same question produces different answers for different datasets**

```
>> Background:
   # Core grounding test. If the engine returns identical answers regardless of which
```

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```
    # files were compared, it is returning a canned LLM response and not reading the data.
200 Given path '/api/v1/ask'                                                                    1
201 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)", "query":    0
        "What changed between these two certification files?" }
202 When method post                                                                           11
203 Then status 200                                                                             0
204 * def answerWithChanges = response.answer                                                   1
206 Given path '/api/v1/ask'                                                                    0
207 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certIdenticalUrl)", "query": "What   3
        changed between these two certification files?" }
208 When method post                                                                            9
209 Then status 200                                                                             0
210 * def answerNoChanges = response.answer                                                     2
211 * assert answerWithChanges != answerNoChanges                                               3
16:10:50.374 ../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-
time-travel-e2e.feature:211
* assert answerWithChanges != answerNoChanges
did not evaluate to 'true': answerWithChanges != answerNoChanges
../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-
e2e.feature:211
```

ask intelligence — ms: 113

Scenario: [8:214] **Count-type query returns an answer containing at least one numeric value**

```
>> Background:
    # "How many new certifications were added?" must return an answer with a digit.
    # "Several certifications were added." without a number is not data-grounded —
    # the engine must read the actual row-count delta and embed it in the response.
218 Given path '/api/v1/ask'                                                                    0
219 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)", "query": "How   0
        many new certifications were added between January and March?" }
220 When method post                                                                            7
221 Then status 200                                                                             0
222 And match response.answer == '#string'                                                     2
223 * assert response.answer.length > 5                                                         1
224 * def containsDigit = function(s){ return s.match(/\d/) != null }                           7
225 * assert containsDigit(response.answer)                                                    91
```

ask intelligence — ms: 12

Scenario: [9:228] **Answer to a factual query is backed by supporting structured evidence**

```
>> Background:
    # Natural language prose alone is not verifiable. The response must include a
    # data field containing structured evidence — technology name, employee count,
    # cert names, old/new values — that backs the claim in the answer string.
232 Given path '/api/v1/ask'                                                                    0
233 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)", "query":    0
        "Which technology had the highest certification growth?" }
234 When method post                                                                            6
235 Then status 200                                                                             0
236 And match response.answer == '#string'                                                     0
237 And match response.data == '#present'                                                      1
238 * assert response.data != null                                                              1
```

ask intelligence — ms: 29

Scenario: [10:241] **Query against identical files answers that no differences exist**

```
>> Background:
    # The NL layer must read the comparison context before answering.
    # "What are the most significant changes?" against a zero-diff context must NOT
    # fabricate differences — it must communicate that the files are unchanged.
245 Given path '/api/v1/ask'                                                                    0
```

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```
246 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certIdenticalUrl)", "query": "What are the most significant certification changes between these files?" }   0
247 When method post   6
248 Then status 200   0
249 And match response.answer == '#string'   0
250 * def answerText = response.answer.toLowerCase()   13
251 * def noChangesInAnswer = answerText.includes('no change') || answerText.includes('identical') || answerText.includes('no difference') || answerText.includes('unchanged')   7
252 * assert noChangesInAnswer   1
16:10:50.550 ../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:252
* assert noChangesInAnswer
did not evaluate to 'true': noChangesInAnswer
../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:252
```

ask intelligence ms: 29

Scenario: [11:255] **Out-of-scope query is gracefully deflected with cannotAnswer flag set to true**

```
>> Background:
# The NL layer must know the boundary of what a certification diff can answer.
# An entirely unrelated question must trigger graceful deflection — not a fabricated
# spreadsheet-data answer. The cannotAnswer flag signals the system understands its scope.
259 Given path '/api/v1/ask'   0
260 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)", "query": "What is the current stock price of Apple Inc?" }   0
261 When method post   19
262 Then status 200   0
263 And match response.answer == '#string'   1
264 And match response.cannotAnswer == '#present'   2
265 * assert response.cannotAnswer == true   1
```

ask intelligence ms: 22

Scenario: [12:268] **Technology-specific query returns an answer that names that technology**

```
>> Background:
# Asking about a specific technology must produce an answer that references it.
# A generic response that never mentions the queried technology is not data-grounded —
# it proves the engine answered without reading the certification domain content.
272 Given path '/api/v1/ask'   0
273 And request { "file1Url": "#(certJan2026Url)", "file2Url": "#(certMar2026Url)", "query": "How did AWS certifications change between January and March?" }   0
274 When method post   12
275 Then status 200   0
276 And match response.answer == '#string'   1
277 * assert response.answer.toLowerCase().includes('aws')   1
```

analyse intelligence ms: 18

Scenario: [13:297] **Cell evolution history contains exactly as many entries as versions submitted**

```
>> Background:
# Each cell's history array must have one entry per submitted version — not more, not fewer.
# A shallow implementation that only diffs adjacent pairs will produce history.length == 2
# for a 3-version submission, failing this check.
301 Given path '/api/v1/analyse'   0
302 And request   1
```

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2026-03-27 04:10:50 PM

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312	When method post	9
313	Then status 200	0
314	And match response.cellEvolutions == '#array'	3
315	* assert response.cellEvolutions.length > 0	1
316	* def correctDepth = karate.filter(response.cellEvolutions, function(c){ return c.history.length == 3 })	1
317	* assert correctDepth.length == response.cellEvolutions.length	0

analyse intelligence ms: 18

Scenario: [14:320] **Versions returned in chronological order regardless of submission order**

>>	Background:	
	# Temporal reasoning is impossible on a misordered sequence.	
	# Submitting versions out of calendar order must be silently corrected by the system —	
	# the response versions array must always be oldest-first regardless of input order.	
324	Given path '/api/v1/analyse'	0
325	And request	0
335	When method post	8
336	Then status 200	0
337	And match response.versions == '#array'	1
338	* assert response.versions.length == 3	1
339	* assert response.versions[0].versionDate <= response.versions[1].versionDate	4
340	* assert response.versions[1].versionDate <= response.versions[2].versionDate	1
341	* assert response.versions[0].label == 'Jan-2026'	1
342	* assert response.versions[2].label == 'Mar-2026'	1

analyse intelligence ms: 33

Scenario: [15:345] **Formula evolution tracks the expression string — not just the computed cell value**

>>	Background:	
	# The most common shortcut is tracking only cell VALUES across time.	
	# A true formula tracker must record the expression string itself	
	# (e.g. =SUM(B2:B18) -> =SUM(B2:B20)) not just the number the formula resolves to.	
	# Each formula history entry must carry a string field starting with "=".	
350	Given path '/api/v1/analyse'	0
351	And request	0
360	When method post	6
361	Then status 200	0
362	And match response.formulaEvolutions == '#array'	0
363	* assert response.formulaEvolutions.length > 0	1
364	* def firstCell = response.formulaEvolutions[0]	1
365	* assert firstCell.history.length >= 2	22

```
16:10:50.705 ../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:365
* assert firstCell.history.length >= 2
js failed:
>>>>
01: firstCell.history.length >= 2
<<<<
org.graalvm.polyglot.PolyglotException: TypeError: Cannot read property "length" from undefined
- <js>.:program(Unnamed:1)

../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:365
```

366	* def firstEntry = firstCell.history[0]	0
367	* match firstEntry.formula == '#string'	0
368	* assert firstEntry.formula.startsWith('=')	0

analyse intelligence ms: 39

Scenario: [16:371] **Point-in-time snapshots for Jan-2026 and Mar-2026 return different states**

>> Background:

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```
    # Can the system reconstruct what the certification data looked like at a specific past date?
    # Jan-2026 and Mar-2026 snapshots must differ because the files are known to have changed.
    # Identical snapshots for all dates means point-in-time reconstruction is not functioning.
375 Given path '/api/v1/analyse'                                                                  0
376 And request                                                                                   0
386 When method post                                                                              8
387 Then status 200                                                                               0
388 And match response.snapshots == '#array'                                                      1
389 * assert response.snapshots.length == 3                                                       2
390 * def snapJan = response.snapshots[0]                                                         1
391 * def snapMar = response.snapshots[2]                                                         2
392 * match snapJan == '#object'                                                                  1
393 * match snapMar == '#object'                                                                  0
394 * assert JSON.stringify(snapJan) != JSON.stringify(snapMar)                                   22
```

analyse intelligence ms: 15

Scenario: [17:397] **Temporal NL answer references a version label or date proving it consulted the timeline**

```
>> Background:
    # "What changed between Jan-2026 and Mar-2026?" must mention "Jan-2026" or "2026-01-31"
    # in the response. A generic LLM answer that ignores the version context entirely
    # cannot pass this test — temporal anchoring is mandatory.
401 Given path '/api/v1/analyse'                                                                  0
402 And request                                                                                   0
413 When method post                                                                              7
414 Then status 200                                                                               0
415 And match response.answer == '#string'                                                        0
416 * assert response.answer.length > 10                                                          1
417 * def answerText = response.answer.toLowerCase()                                              2
418 * def referencesVersions = answerText.includes('jan-2026') || answerText.includes('mar-2026') || response.answer.includes('2026-01-31') || response.answer.includes('2026-03-20')   1
419 * assert referencesVersions                                                                    0
```

analyse intelligence ms: 17

Scenario: [18:422] **Analyse without a query returns full structured breakdown and no answer field**

```
>> Background:
    # query is optional. Without it the endpoint must still return the complete structural
    # breakdown — versions, cellEvolutions, formulaEvolutions, snapshots.
    # The answer field must be absent entirely when no query was provided —
    # not null, not an empty string, but #notpresent.
427 Given path '/api/v1/analyse'                                                                  0
428 And request                                                                                   0
437 When method post                                                                              10
438 Then status 200                                                                               0
439 And match response.versions == '#array'                                                       1
440 And match response.cellEvolutions == '#array'                                                 0
441 And match response.formulaEvolutions == '#array'                                              0
442 And match response.snapshots == '#array'                                                      0
443 And match response.answer == '#notpresent'                                                    2
16:10:50.793 ../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:443
And match response.answer == '#notpresent'
match failed: EQUALS
  $ | present (NULL:STRING)
  null
```

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```
  '#notpresent'

../../../../problems/ai-excel-time-traveller_1774096301241/feature/excel-version-comparison-time-travel-e2e.feature:443
```

analyse intelligence ms: 22

Scenario: [19:446] **New technology domains emerging in later versions are reflected in the NL answer**

```
>> Background:
   # The temporal engine must detect structural additions — not just value changes.
   # If Mar-2026 introduces Kubernetes certifications absent in Jan-2026, the NL answer
   # to a "what's new?" query must acknowledge that emergence.
   # A "no change" style answer here proves the system is not reading the diff.
451 Given path '/api/v1/analyse'                                                    0
452 And request                                                                     0
462 When method post                                                                14
463 Then status 200                                                                 0
464 And match response.answer == '#string'                                          1
465 * assert response.answer.length > 10                                            0
466 * def answerText = response.answer.toLowerCase()                                0
467 * def claimsNoChange = answerText.includes('no new') || answerText.includes('no change')   2
      || answerText.includes('identical') || answerText.includes('no difference')
468 * assert !claimsNoChange                                                        1
```